

Building a unified science of sensorimotor control with Densely Overlapping Measurement Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

Build the DOME — a new class of scientific instrument that records the complete perception–action loop of a behaving agent as a single calibrated, uncertainty-tagged measurement — and the autonomous organization required to turn it into shared infrastructure, fusing four disconnected sciences of behavior into one.

1.1. The Observable

- Every agent, living or engineered, runs the same loop: it senses a thin slice of its environment through imperfect transducers and pushes back against the world to move toward a goal.
 - Information flows in, forces flow out; the brain exists to move the bones.
 - This loop is the shared object of four fields that rarely speak: perceptual and motor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI.
- We can already measure each thread of the loop with extraordinary precision — but only one thread at a time.
 - Silicon probes resolve single neurons; markerless capture and force plates resolve kinematics and kinetics; electromyography resolves individual motor units; mobile eye trackers resolve gaze.
 - Each instrument is superb in isolation and blind to the rest of the loop.
- No one can record the whole loop at once, because the integration is harder than any single lab can sustain.
 - Every so often a team stitches several threads together for one heroic study.
 - But the method lives only in a Methods section, and the wiring decays on the timescale of a single PhD, as the burden of maintaining the integration outstrips what an academic lab can hold together.
 - Each of those teams was reaching for the same measurement — the coupled record of perception **and** action — and each was stopped by the same wall.
- The unmet need is therefore a single instrument, not a better sensor.
 - It must capture every empirically available channel of an agent's perception and action at once: spatially calibrated, temporally synchronized, and rendered commensurable across labs and species.
 - It must be faithful enough that its record can predict neural activity along the sensory and motor pathways — the fidelity bar that separates a recording from a measurement.

1.2. The Instrument

- We call this instrument the DOME — a Densely Overlapping Measurement Environment.
 - It is a densely instrumented region of real space engineered to record every measurable channel of the agent–environment interaction at once: full-body kinematics and kinetics, muscle and motor-unit activity, binocular gaze and reconstructed retinal input, and, where feasible, neural activity.
 - It is the inverse of a space telescope: a room of lenses aimed inward, at the place where the body meets the world.

- The DOME is one instrument built in three forms, each defined by the role it plays in the network.
 - DOME-L is the warehouse-scale flagship in greater Boston: it pushes each instrument to its limits, and is large enough to physically contain and cross-validate the smaller forms, which makes it the metrological reference for the whole network.
 - DOME-S is the lab- and classroom-scale form we disseminate to other groups, extending the webcam capture volumes the FreeMoCap Foundation already supports worldwide.
 - DOME-Mobile is the wearable form we carry outside the lab, used to validate lab-bound research against real-world behavior in natural terrain.
 - Every form produces the same kind of record, because the instrument is defined by its measurement ontology rather than by any single sensor: a camera-based and an inertial estimate describe the same kinematics and fuse into one estimate better than either alone.
- The DOME is not built around one new sensor; we advance the measurement on both halves of the loop, so the platform's own capability frontier keeps moving and the instrument tells us which transducer to build next.
 - On the motor side, we build an actuated, self-calibrating camera array: daisy-chainable Power-over-Ethernet cameras with swappable sensors on programmable mounts, re-aimable and re-focusable from a single console while staying metrically calibrated as one instrument – solving the lived problem of large capture volumes whose cameras get frozen in place because re-aiming and re-calibrating them by hand is too costly.
 - We fuse these outside-in camera estimates with inside-out inertial estimates: cameras are accurate but noisy, inertial capture is precise but drifts, and their principled fusion carries explicit uncertainty rather than hidden guesswork, making trustworthy inverse dynamics and muscle-force estimation tractable from markerless data.
 - On the sensory side, we build the flagship new transducer: a binocular mobile eye tracker that measures degrees of freedom commercial eye trackers do not – ocular torsion and lens accommodation – because a faithful reconstruction of what lands on a moving agent's retina requires them, and because smartphone-grade imagers behind controlled illumination make this a camera-quality bet, not a physics bet.
- We have already built a working animal-scale version of this instrument.
 - The team's ferret and mouse eye trackers already integrate three-camera gaze, full-body kinematics, and world cameras in one calibrated system – a working animal-scale instance of the flagship instrument – and the human, mouse, and ferret builds are being made functionally equivalent so the same measurement crosses species.
- [ALLUDE → §3: DOME-L warehouse specs, DOME-S build plans and FreeMoCap community metrics, DOME-Mobile sensor suite, eye-tracker DoF targets and validation, actuated camera array engineering, camera↔IMU fusion uncertainty budgets, force-plate floor panels, ARGpV3, drone swarm (later phase).]

1.3. The New Capability

- Because every channel is spatially calibrated, temporally synchronized, and expressed in one sensor-grounded ontology, DOME records make heterogeneous sensor streams directly commensurable, with uncertainty traced from each transducer to every derived value.
- This structure makes DOME data consumable by modern reinforcement-learning and robotics stacks with minimal reshaping.
 - A measured bone segment and a simulated robot link are the same RigidBody; a measured ground reaction force and a simulated contact force are the same quantity.

- ▶ Embodied AI and legged robots gain real-world sensorimotor corpora annotated with the explicit uncertainty bounds that safe learning requires.
- The same calibrated array provides physical ground truth for computer vision at large scale.
 - ▶ Back-projecting the reconstructed 3D solution onto each camera's 2D view yields a reprojection-error signal that directly improves the underlying skeleton trackers, both as single-camera models and as in-volume estimators — an instrument, in the Topic's own language, engineered to produce better AI training data.
- The archive appreciates over time, because the ontology defines slots for channels no current sensor fills.
 - ▶ Once a tracker measures torsion or accommodation, models trained on the new channels can back-predict them for older recordings made before any instrument could capture them, and the same principle up-fills data from lower-cost DOMEs using models trained on flagship data.
- The instrument closes the hypothesis loop on itself.
 - ▶ Inverse reinforcement learning on DOME records extracts the reward a biological agent appears to optimize, which both jump-starts robot training and becomes a testable hypothesis about the underlying neural computation.
 - ▶ That hypothesis is then probed back inside the DOME by controlled perturbation of the loop, through a reconfigurable augmented-reality ground plane of LED floor panels, projection, and VR that reshapes the terrain a real agent navigates while it is measured.
 - ▶ Observation, modeling, and intervention become a single apparatus: the full hypothetico-deductive cycle inside one instrument.
- [ALLUDE → §3: reprojection-error pipeline, robot RL integration with Isaac Lab/MuJoCo, IRL methodology and validation, backfill/upfill statistical framework, ARGpV3 build plan and lineage from ARGpV1 (Matthis 2013–2017), environment-manipulation modalities, the five-station closed loop.]

1.4. Why Unmet

- An instrument like this cannot be built inside the institutions that most need it, for structural reasons that are well understood but rarely stated plainly.
 - ▶ A trainee's job is to learn by doing science; a tool's job is to encode hard-won mastery so others need not re-learn it — and these point in opposite directions.
 - ▶ Academia rewards the experienced builder for novel findings and pushes infrastructure onto trainees who rotate out on degree timelines, so the shared instrument never accumulates the mastery that would make it trustworthy.
 - ▶ The result is the well-documented graveyard of abandoned research software: the incentive structure rewards the thread, not the loom.
- A second structural problem is scope.
 - ▶ A tool defined by a research domain partitions its users into specialists; a tool defined by a measurement unites everyone who needs that measurement, however different their goals, and becomes a boundary object — a commons — precisely because it imposes none of them.
 - ▶ The FreeMoCap Foundation already ran this experiment: scoped entirely to the measurement, refusing to claim any domain, it became a commons where biomechanists, neuroscientists, roboticists, animators, and game designers now cross-pollinate across 15,000 users in 152 countries.

- ▶ OpenCap and DeepLabCut are excellent and, by being domain-scoped, are the foils that prove the point: measurement-not-domain scoping is what creates a commons rather than a user base.
- A third problem is organizational shape, and it names the answer the Observable left open: the measurement needs its own organization.
 - ▶ Conway's Law holds that organizations produce systems mirroring their own structure, so an enterprise of domain-named departments, journals, and grants produces domain-scoped instruments, reviewed by reviewers who can only score work inside one domain — never the long, unglamorous metrology that turns a proof of concept into a tool others trust.
 - ▶ The PI left a tenure-track faculty position because the institution could not support this work, and founded the FreeMoCap Foundation as a 501(c)(3) whose primary output is open scientific instruments.
 - ▶ The X-Lab's operating model follows: scope the deliverable to the instrument, and fan the science out to a standing network of collaborators — human perception-and-action researchers, robotics and prosthetics groups, and animal labs spanning mouse, ferret, and guinea fowl — who use the tools and convene to feed insight back.
 - ▶ Building the tool and leaving the science to others is the mechanism of impact, not a retreat from it: a working shared instrument as the success metric, and full-time career engineers held for years, are exactly the X-Labs model and exactly what a university cannot provide.
- [ALLUDE → §4, §5: PI's expertise and departure from Northeastern, EI's enterprise software background, AC's clinical validation dissertation, KM's mobile DOME lineage, MN's large-scale lab expertise, JKL's AI systems role, RR's organizational health and NSF grant management, NR's CEO operations role, collaborator network roster, governance (FMC-X within FMC-F, PI has final say), autonomy (already autonomous at submission), semi-annual congress/hackathon/workshop/conference cadence, community grants program, FreeMoCap scale receipts.]

1.5. The Vision

- The end-state is a single empirical science of sensorimotor control.
 - ▶ Perceptuomotor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI — today four parallel literatures with incompatible tools and vocabularies — become commensurable, so a finding in one becomes evidence in another and four bodies of work collapse into one cumulative science of real-world behavior.
 - ▶ On the very same measurements sits a second end-state: the uncertainty-tagged corpora embodied AI and robotics need to learn real-world sensorimotor behavior, reshaping a field of science and seeding a sector of technology at once.
- In seven years, the jogger crossing a rocky field is a routine measurement.
 - ▶ Her kinematics, forces, muscle activity, gaze, the terrain beneath her, and the image on her retina are recorded as one object, with a defensible uncertainty on every value, by an instrument a graduate student can operate.
 - ▶ The same kind of record — lower precision, not a different kind — is produced by a classroom in Ohio and by a ferret hunting in a laboratory, and the three can be compared because every value traces to the same references.
- We will build the full range and put it everywhere.

- ▶ Hundred-dollar three-camera kits for students, mobile eye trackers that measure ocular torsion and accommodation, field-deployable gait laboratories, and wearable research environments — all interoperable, all producing the same kind of record.
- ▶ Deployed in hospitals and physical-therapy clinics, athletic centers and robotics labs, classrooms, and on the International Space Station — each one an instance of the same instrument, the same measurement, the same growing archive.
- The telescope showed us our place in the universe by pointing outward; the DOME shows us ourselves by pointing inward, at the place where the body meets the world — bringing the ordinary condition of behavior, an agent using vision to place its feet in an unconstrained environment, into routine, traceable measurement.



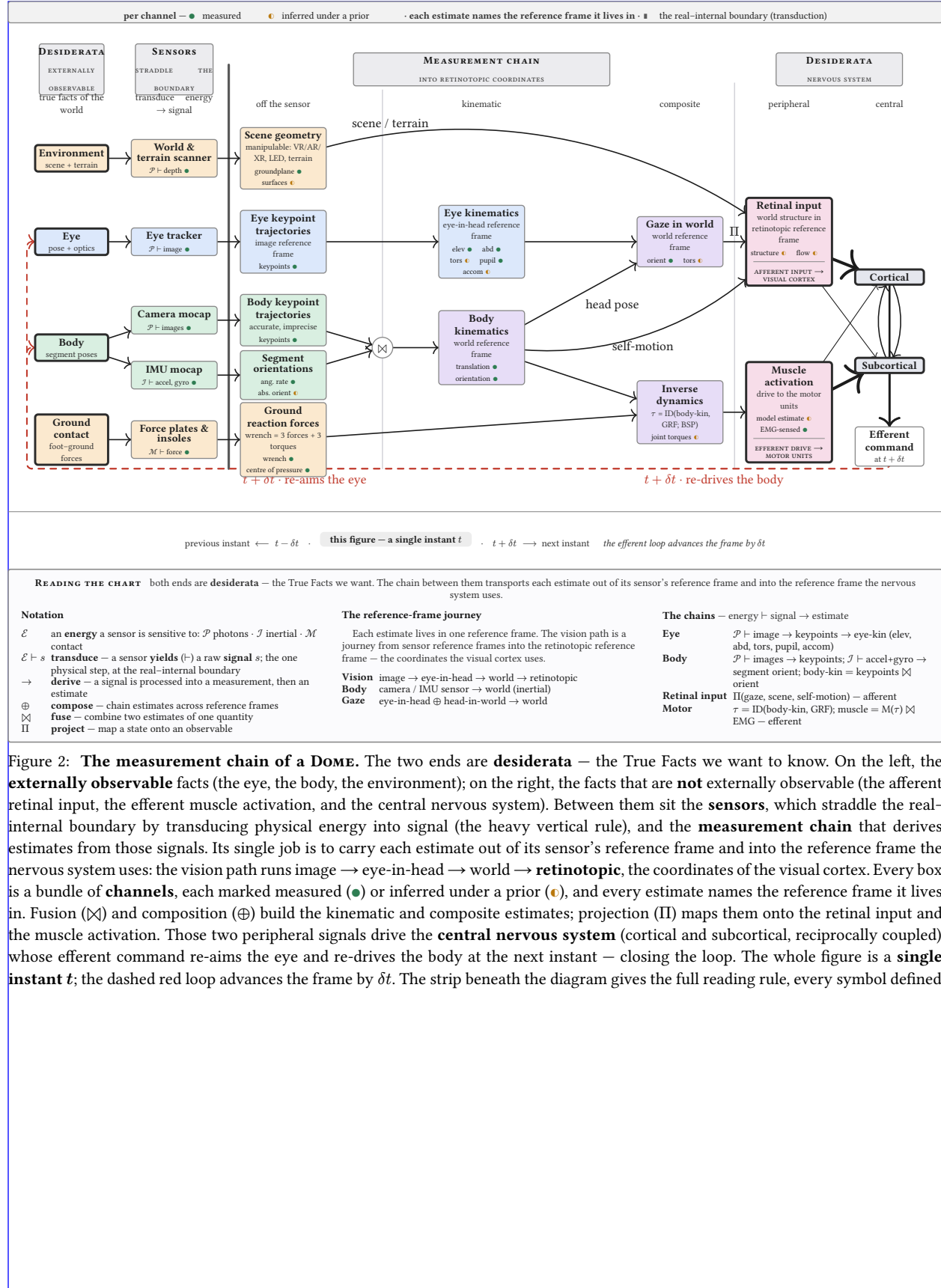


Figure 2: **The measurement chain of a DOME.** The two ends are **desiderata** — the True Facts we want to know. On the left, the **externally observable** facts (the eye, the body, the environment); on the right, the facts that are **not** externally observable (the afferent retinal input, the efferent muscle activation, and the central nervous system). Between them sit the **sensors**, which straddle the real-internal boundary by transducing physical energy into signal (the heavy vertical rule), and the **measurement chain** that derives estimates from those signals. Its single job is to carry each estimate out of its sensor's reference frame and into the reference frame the nervous system uses: the vision path runs image → eye-in-head → world → **retinotopic**, the coordinates of the visual cortex. Every box is a bundle of **channels**, each marked measured (●) or inferred under a prior (●), and every estimate names the reference frame it lives in. Fusion (⊗) and composition (⊕) build the kinematic and composite estimates; projection (II) maps them onto the retinal input and the muscle activation. Those two peripheral signals drive the **central nervous system** (cortical and subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant t** ; the dashed red loop advances the frame by δt . The strip beneath the diagram gives the full reading rule, every symbol defined.

2. Technology Landscape

2.1. The measurement threads and their instruments

- The DOME unifies six measurement threads, each served by a mature but isolated instrument ecosystem.
 - Kinematics: markerless pose estimation (DeepLabCut, SLEAP, VitPose, FreeMoCap), marker-based (Vicon, Qualisys, OptiTrack), and inertial (Xsens, Notch), with a gap between clinically-valid kinematics and inverse-dynamics-grade accuracy.
 - Kinetics: force plates (AMTI, Bertec), pressure-sensing insoles, and instrumented treadmills, with ground reaction forces still hard to measure outside the lab.
 - Eye tracking: Pupil Labs (Core, Neon, Invisible), Tobii, and SMI, none of which capture torsion or accommodation.
 - Neural recording: Neuropixels, miniscopes (1-photon calcium imaging), and mobile EEG, with neural activity still hard to record during unconstrained movement.
 - Muscle activity: surface and intramuscular EMG at motor-unit resolution, with a gap between what OpenSim models need and what current sensors deliver.
 - Environment: photogrammetry, Gaussian splatting, neural radiance fields, RGB-D (stereo, LiDAR, structured IR), and event cameras.

2.2. The composition problem

- These instruments do not compose, for four concrete reasons.
 - Coordinate frames are incompatible, because each sensor defines its own.
 - Clocks are unsynchronized, because there is no common timebase.
 - Semantic schemes are non-interoperable: a “gaze point” in one system is not the same quantity as in another.
 - Uncertainty does not propagate across modality boundaries.
- Mobile Brain/Body Imaging (Makeig et al. 2009) is the fifteen-year case study: a methodology, not a platform, because the integration burden falls on each lab.

2.3. The metrology gap

- The field co-records without co-measuring.
 - Calibration quality $\kappa = (\kappa_s, \kappa_t)$ is the fundamental bottleneck. [Reference observability calculus.]
 - The GUM/VIM framework applied to multimodal capture shows why an unbroken chain of traceable calibrations matters — and why no existing system provides one.

2.4. The ontology gap

- Existing tools are sensor-specific rather than measurement-specific.
 - They are built around a particular camera model or force-plate vendor, not around the measurement itself; kinematics is kinematics whether it came from cameras or IMUs.
 - The consequence: data from different labs, species, and equipment generations cannot be compared, pooled, or cumulatively improved.
 - The DOME’s sensor-grounded ontology is the solution: define the measurement, not the sensor.

2.5. The CV field’s plateau

- Computer-vision pose estimation has drifted away from the pixels toward ungroundable targets.

- Targets such as 3D-from-2D, SMPL mesh fitting, and semantic segmentation lack physical ground truth.
- Trackers still benchmark against aging COCO data scraped from early-2000s Flickr, and fail on out-of-distribution movement – clinical populations, extreme athletics, animal locomotion.
- The field needs physical ground truth at scale, which a calibrated multi-camera array provides, along with corpora of the movements it currently cannot track.

2.6. The robot training bottleneck

- Reinforcement learning for legged robots is limited by the sim-to-real gap, not by simulation.
 - Simulation has advanced rapidly (MuJoCo, Isaac Lab, Genesis), but sim-to-real transfer remains the critical barrier.
 - The missing resource is ground-truth, uncertainty-tagged, multimodal recordings of real agents in real environments.
 - Existing datasets (Human3.6M, AMASS, GRAB) are single-modality, lab-constrained, or lack uncertainty quantification.

2.7. Quantitative landscape figure

- A single figure will locate current capability against the DOME’s target.
 - For each modality, plot current commercial and academic capability against the DOME’s target, with the “uninstrumented region” highlighted.
 - Axes: spatial precision, temporal synchronization quality, degrees of freedom captured, and usability/deployability.

3. Outcomes

- Develop, validate, and disseminate - DOME Variants
 - **DOME-L** — the flagship instrumented volume, in the greater Boston area. Large enough to fully enclose the smaller variants for validation and metrological grounding.
 - **DOME-S** — the extension of the commodity-camera capture volumes already built and disseminated through the FreeMoCap Project. Representative of the DOME a research lab or a classroom would build.
 - **DOME-Mobile** — a wearable, self-contained DOME producing Dense Observations in unconstrained indoor and outdoor environments, continuing the lineage of the PI's gaze/gait and retinal optic flow work¹⁻⁴.
 - Validate in large DOME-L so we can trust the outdoors
 - Also - disseminate custom use-directed builds of DOME variants through network of collaborators studying Humans, animals, and robots.

3.1. Bottlenecks

- **Technology**
 - **New eye tracker**
 - Best ever made
 - Add every sensor we can, see which allow us to measure DoFs we need, optionally scale back and remove
 - Optionally also make cheaper version for classrooms - fully compatible data output (with less precision and depth)
 - Remake the field by retraining the next generation of researcher in the tools of the field you want to build
 - Dense instrumentation in the eye and world cameras
 - **Eye sensor array**
 - includes
 - High res IRGB
 - High speed IRGB
 - MEMS single-pixel sensor
 - LiDAR scanner
 - IR/Structured IR illuminators
 - Event Camera
 - **World sensor array**
 - Build different form factors (big heavy, small light)
 - Head- or Drone- mountable
 - Stereo RGB, Structured IR, LiDAR, etc
 - High density multi-model environment scan
 - Fast scan (RGB-D)
 - Slow scan (Textured Mesh)
 - Semantic segmentation (via SAM3, YOLO, etc)
- **Auto-configuring Mocap Camera Array**
 - Large scale mocap too hard to control, cuts against usefulness of large volumes. Legit barrier to progress and usefulness of mocap!!

- ▶ Our Auto-calibrating, Programmatically controlled Mocap Camera array will help!
 - Allow centrally controlled definition of capture volume within available range
 - Auto-configure for high density tracking within a small space, or more diffuse tracking across a larger space. Higher ceiling for drones, or groundplane focus for locomotion
 - Use to cross-validate smaller DOMEs within its space, Programmatically explore best practices for camera placement and calibration procedures
- ▶ **Each camera:**
 - Fully actuated Extrinsics (pitch, tilt, roll motor) and Intrinsics (Zoom, aperture, focus)
 - **Power Over Ethernet** (daisy-chainable power + data, so we can control arbitrary cameras from a single wire coming off the computer)
 - Modular:
 - Sensor (high def, high speed, spectral sensitivity, size, etc)
 - Illumination (Matchin spectrum of sensor)
 - Lens Assembly (zoom, aperture, etc)
- **Hybrid Body kinematics**
 - ▶ Fuse IMU- kinematics iwth Camera-kinematics for BEST kinematics (able to do inverse dynamics (note - also needs force plates))
- **Drone swarm mocap**
 - ▶ Do it in comparison of IMU- mocap to static DOME cameras, then itnegrate a **Swarm of Drone-mounted World-Sensor arrays** that ground inside-out estiamtes of IMU- and head-mounted World-sensor array during unconstrained outdoor used (train drones and develop swarm controller by tracking them within large DOME-L)
 - ▶ Drone kinematics will be noisy, but fusable with IMu-mocap for grouned outdoor data
- **Modular Force Plate floor panels**
 - ▶ Floor is all removable panels
 - ▶ Integrate force plates at key points
 - ▶ Create **Terrain** and **LED sceen** panels that can attached to top of force plates, give force reading under defined 3d terrain and interactive LED panels (for cntrolled experiments that manipulate location and texture motion of target footholds)
 - ▶ Each 0.5m square of the ground can be: Forceplate, Terrain/LED panel, or both (force plate stacked with terrain that is either inert or composed of LED panels)
 - ▶ Force plates allow measuring GRF and downstream OpenSim and Inverse Dynamics stuff to estimate muscle activity
 - ▶ Terrain allows validation of head and drone mounted world scanners
 - ▶ LED panels allow controlled experiments of visual/locomotor control theory, and useful human DOME data to use Inverse Reinforcement Learning to extract apparent human control policy, and feed that policy AND and the actual DOME data into robot RL control systems solvers

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3.2. Timeline		
NOTE - THIS IS A PLACEHOLDER!!!! THIS IS A PLACEHOLDER		
Phase	Name	Description
1	Organize	Set up Organization (Phase 0) – Solidify release of FreeMoCap v2
2	Architect	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	Execute	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	Stabilize	Establish OODA loop within expected standard operations
5	Expand	Once stabilized, expand key performance metrics
6	Align	Focus on building deeply integrated community of users and contributors (Transition BDFL → BDFaSPoT)
7	Establish	Focus on long term stability and healthy growth
8	Maintain	Beyond proposal timeline – Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

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Table 1: Timeline of Phased Milestones		
NOTE - THIS IS A PLACEHOLDER!!!! THIS IS A PLACEHOLDER		

NOTE - Names abbreviated in public draft. Check initials to real names in local .env (or request it, if you need it)

Name / Role	Qualifications
Jonathan Matthis, PhD President	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics¹⁻³
NR, CEO	• Expertise in technically and software dev • Previous CEO XP, founded and ran own buisiness • CEO helps PI apply his specific expertise to the core tasks of the XLab
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design and brain imaging • Worked with PI on core FMC dev and validation
JKL, PhD CAIO	• PhD - Cognitive Science • Director - Cognitive Science Program • Current - Senior AI Applications Engineer @ SOLID • Expertise in use of AI in classroom, lab, and enterprise
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship (founded own Bookkeeping coop) • Taught Finance Theory at university level
KM Project Manager (?)	• Key driver of³ • Worked with PI on core tech² • Expertise in key technologies
MN Project Manager (?)	• Expertise in clinical biomechanics, • Medical systems design, • Mechatronics for clinical tools, • Multi-modal motion capture lab management

Table 2: Senior/Key Personnel Qualifications

4. Team Capabilities Statement

4.1. Senior Team Capabilities

- Jon - done it before in humans, currently doing it in ferrets and mice. Left tenure track position to run FreeMoCap, XLab funding will accelerate that mission
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development
- NR - XP as CEO and working in tech industry. Hiring her as CEO allows Jon/PI to focus on applying their domain expertise on the core work of the lab
- JKL - AI expertise in teaching, research, and industry. Will build internal AI systems for core software, developer, user, and student assistance.
- AC - focusing clinical adoption and bench-to-bedside transitions. Wrote his dissertation on validation of freemocap. Has all skillsets needed for DOME-S
- MN - XP in mechatronics and large-scale lab building and management. Has core skillset needed for DOME-L
- KM - Part of original laser skeleton team, has all necessary sub-skills to build core instrumentation of Mobile DOME

4.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS, Natural Bx, VR/AR and eye tracking
KB	Human/NHP	Visual Neuroscience, Optometry
JY	Human/NHP	Computational neuroscience, Eye tracking, Ephys
GD	Human	Eye tracking, VR/AR, Perceptual/Motor control
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH, SH	Robotics, Prosthetics, Exoskeletons	Control theory

Table 3: Collaborator Network

4.3. Governance

- FMC-X will exist as a independent project within FMC-F organization
- FMC-F will handle other work (e.g. FreeMoCap Project), but otherwise the missions almost fully overlap
- Nearly identical leadership between FMC-F and FMC-X, with final say on all topics falling to the President (who is also the PI) ensures fast decision making (no higher management to talk to - PI is the top of the org chart)
- Collaborator network and user community consulted for design decisions, work targets, and roadmap
 - Handled via “Request for Comments” style announcements (like Python’s PEP), so progress can’t be stalled by community problems.

- Annual Workshop/Hackathon/Conference/Congress to bring together collaborator network to discuss progress, tech targets, friction, etc.
 - Workshops to train students and out-of-domain experts
 - Hackathons to on-board developers and foster in-person connections between normally remote workers (important for FOSS)
 - Conferences to discuss progress and research done with DOME tech
 - Congress to discuss org status, plans, and future (annual State of the Skelly address)
 - Developer fund, Community Grants Program, etc to support FOSS network
 - Use Community Grants Program to test different strategies for community engagement and consensus taking (Eurovision style voting, DAO methods, etc)

4.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
 - This project will happen no matter what, but with XLabs funding, we could truly change the world

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